Predicting Content Demand and Dynamic Pricing with Climate-Driven Analytics

Project Overview

What is Variable Pricing?
Variable pricing refers to the strategy of adjusting content prices based on shifting demand signals, such as the time of day, user behavior, or external factors like weather conditions. Unlike fixed pricing, it allows platforms to respond flexibly to conditions that influence user interest, engagement, or willingness to pay.

YAKWETU™ Online Ltd, via its MyMovies.Africa™ platform, distributes African content globally. As the platform grows, strategic decisions around pricing and content engagement must be increasingly data-driven. Global evidence suggests that weather patterns can influence user mood and content consumption behavior.

This project proposes using climate (weather) data in conjunction with historical viewing and transaction data to analyze viewing behavior and predict periods of high engagement. These insights will inform the design of variable pricing models that align with user demand patterns, ultimately supporting more personalized experiences and monetization strategies.

Project Goals

- Integrate historical weather data with platform usage data
- Identify correlations between weather conditions and content engagement
- Predict high-demand and low-demand periods using machine learning
- Simulate pricing strategies based on forecasted conditions
- Provide actionable recommendations for content scheduling and pricing

Key Features

Feature	Description
Weather + Usage Correlation	Discover how weather affects content views, completion, and transactions
Genre-Level Analysis	Identify which genres spike during certain weather conditions
Predictive Modeling	Use ML to forecast viewership likelihood based on climate features
Variable Pricing Simulation	Model how dynamic pricing could boost revenue on high-demand days

Technical Overview

- **Data Sources:**
    - Weather: OpenWeather API, historical climate datasets
    - Usage: Existing internal databases for views, transactions, and completions
- **Tools:**
    - Python (pandas, scikit-learn, seaborn), SQL, Power BI
- **Architecture:**
    - Merge weather + usage datasets

- Train predictive models (regression, classification)
- Simulate pricing scenarios using model outputs

Stakeholders & Use Cases

Stakeholder	Use Case
Product Team	Tailor recommendations by time-of-day and local weather
Marketing	Run weather-triggered promotions or bundle campaigns
Content Team	Forecast content demand and genre popularity
Executives	Use predictive demand to plan business strategy and pricing
Data Team	Ensure that weather and behavioral data integration is robust

Deliverables for Student Team

Phase	Deliverable
Discovery	Literature review and EDA of weather + viewing data
Setup	Source weather data and clean platform logs
Modeling	Develop a predictive model to forecast content demand
Simulation	Test pricing simulations under different weather scenarios
Handover	Present findings, model results, documentation, and strategic next steps